

5. Data Preparation and Exploratory Data Analysis

5.1 Objective of the Exploratory Data Analysis

Before training machine learning models, it is necessary to understand the statistical properties and structure of the dataset. Exploratory Data Analysis (EDA) plays a critical role in identifying patterns, detecting anomalies, and evaluating the suitability of the dataset for machine learning tasks.

In the context of music genre classification, the dataset consists of numerical representations of audio recordings extracted using signal processing techniques. These features capture temporal, spectral, cepstral, and harmonic properties of music signals.

The objective of this analysis is to:

- verify the structure and quality of the dataset,
- analyse statistical distributions of audio features,
- identify correlations between features,
- detect potential outliers and anomalies,
- assess dataset limitations and potential bias,
- derive conclusions that influence the experimental design of the classification system.

5.2 Dataset Structure and Integrity Analysis

The dataset used in this project is derived from the **GTZAN Music Genre Dataset**, a widely used benchmark dataset in music information retrieval research. The dataset contains audio recordings belonging to ten musical genres. Each genre contains approximately 100 tracks, resulting in a total of around 1000 audio files, each with a duration of 30 seconds. The audio files are stored in .wav format, ensuring consistent audio quality and compatibility with signal processing libraries.

The dataset includes the following genres:

- blues
- classical
- country
- disco
- hiphop
- jazz
- metal
- pop
- reggae
- rock

Feature extraction was performed using the Librosa library, which generated a structured tabular dataset where each row corresponds to one audio recording and each column represents a numerical audio descriptor.

The extracted features include:

- Tempo
- Zero Crossing Rate (mean and standard deviation)
- RMS Energy (mean and standard deviation)
- Spectral centroid (mean and standard deviation)
- Spectral bandwidth (mean and standard deviation)
- Spectral rolloff (mean and standard deviation)
- MFCC coefficients (1–13 mean and standard deviation)
- Chroma features (1–12 mean and standard deviation)

In total, the dataset contains over 60 numerical features and one categorical variable representing the music genre.

Data integrity verification confirmed that:

- no missing values were present in the dataset,
- all audio files were successfully processed,
- the dataset contains a consistent structure.

These observations indicate that the dataset is suitable for further analysis and machine learning experiments.

5.3 Genre Distribution Analysis

Understanding the distribution of classes is important for evaluating whether the dataset is balanced.

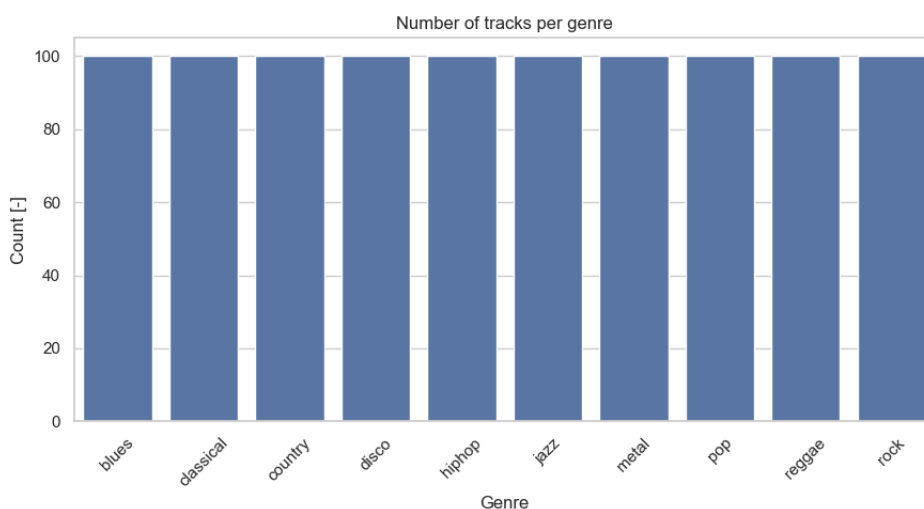


Figure 1 Number of tracks per genre

The class distribution plot confirms that the dataset is balanced across genres. Each genre contains approximately the same number of tracks. Balanced class distribution is beneficial for supervised classification tasks because it reduces the risk of model bias toward specific classes.

As a result, no additional resampling techniques such as oversampling or undersampling are required during model training.

5.4 Distribution of Temporal and Energy Features

The next step of the analysis examines the distributions of temporal and energy-related audio features:

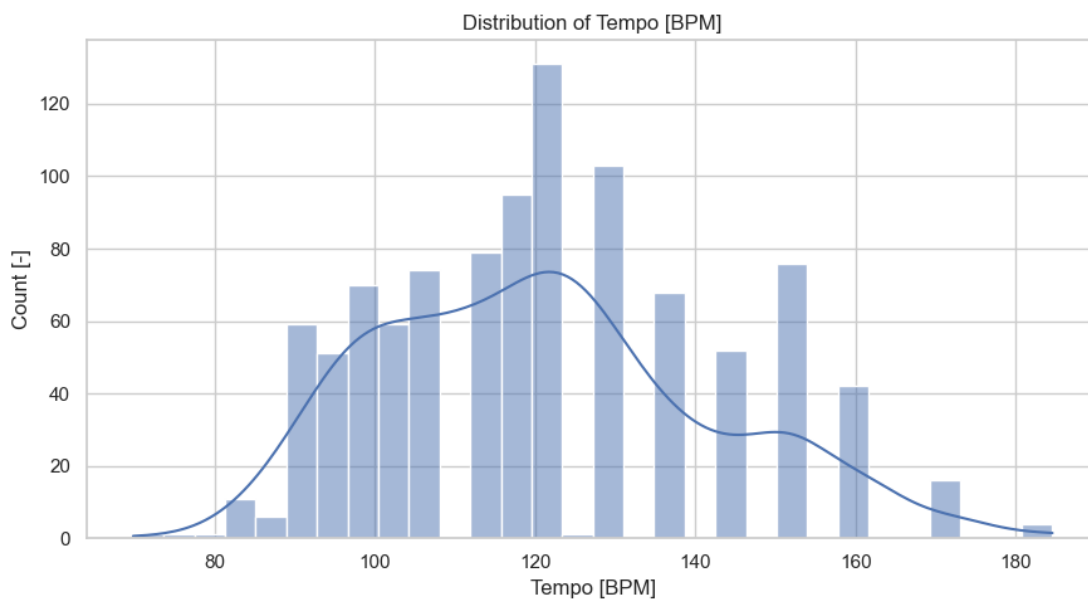


Figure 2 Distribution of tempo

The tempo distribution ranges approximately between 70 and 180 beats per minute. This wide range reflects the rhythmic diversity across different musical genres. Genres such as classical and jazz typically exhibit slower tempos, whereas rock and electronic music often contain faster rhythmic structures.

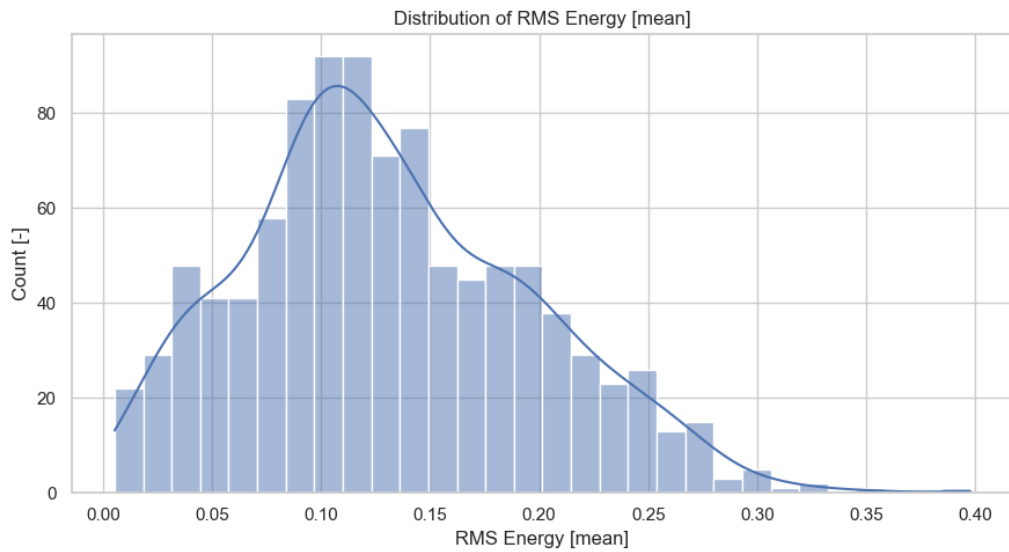


Figure 3 Distribution of RMS Energy (mean)

RMS energy represents the overall loudness of the signal. The distribution shows that most tracks fall within moderate energy ranges, indicating similar recording loudness levels across genres.

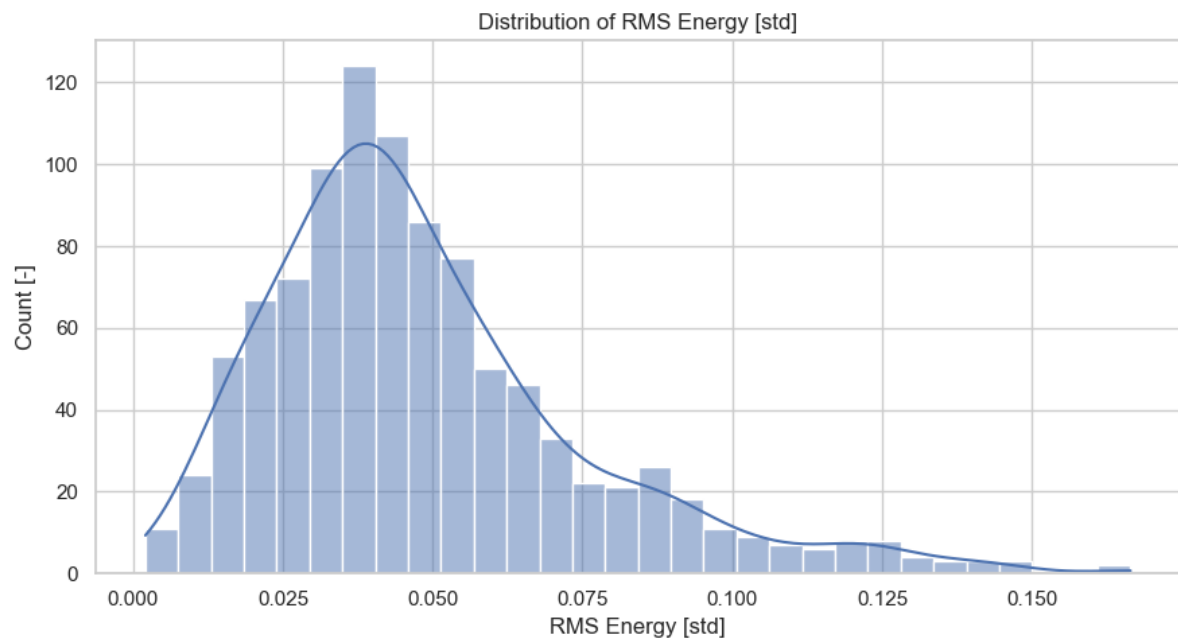


Figure 4 Distribution of RMS Energy (std)

The standard deviation of RMS energy reflects variability in signal intensity throughout the track. Higher RMS standard deviation values suggest greater dynamic variation in the audio signal.

5.5 Spectral Feature Distributions

Spectral features describe the distribution of frequencies within the audio signal and are commonly used in music information retrieval tasks.

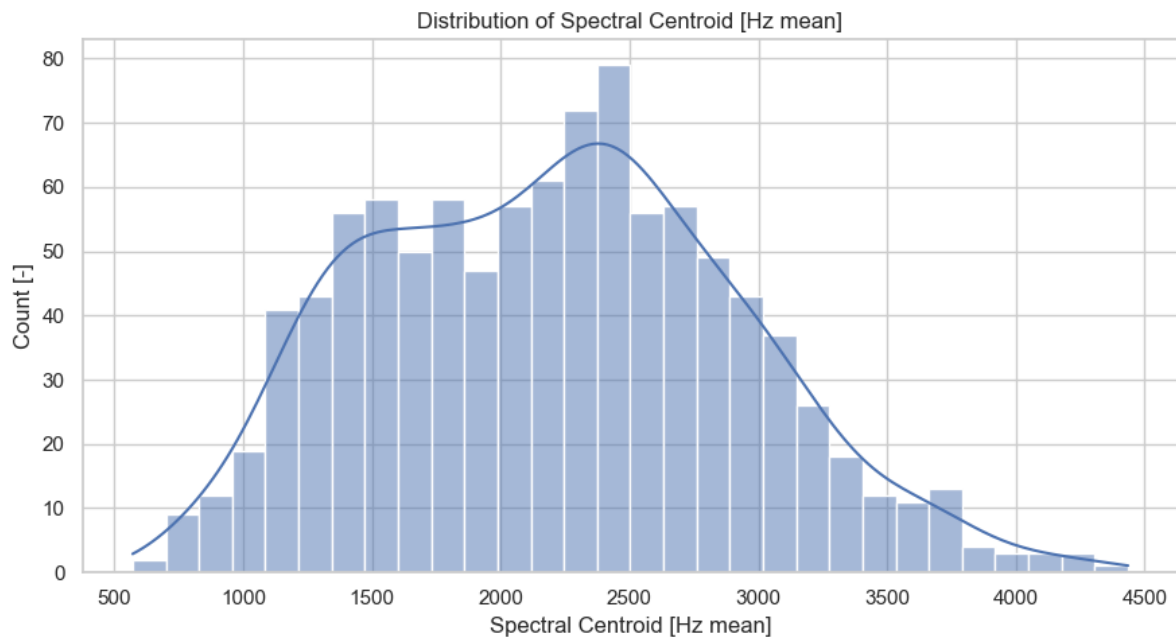


Figure 5 Distribution of Spectral Centroid (mean)

Spectral centroid is often interpreted as the “brightness” of the sound. Higher values correspond to signals containing more high-frequency energy.

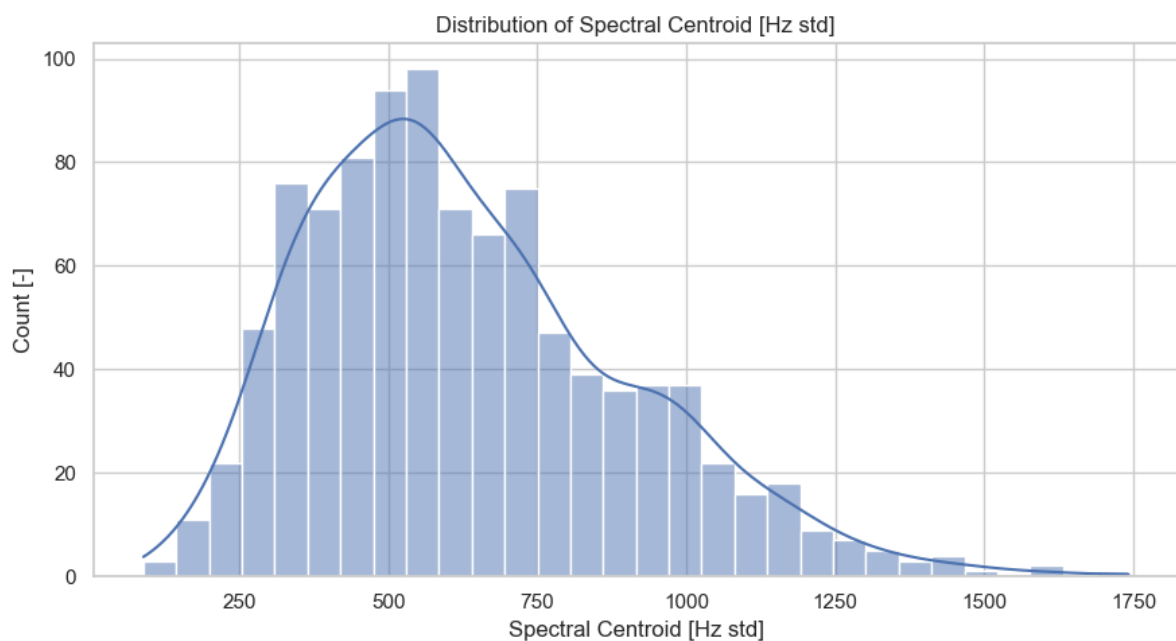


Figure 6 Distribution of Spectral Centroid (std)

The variation of spectral centroid across tracks indicates differences in timbral characteristics between genres.

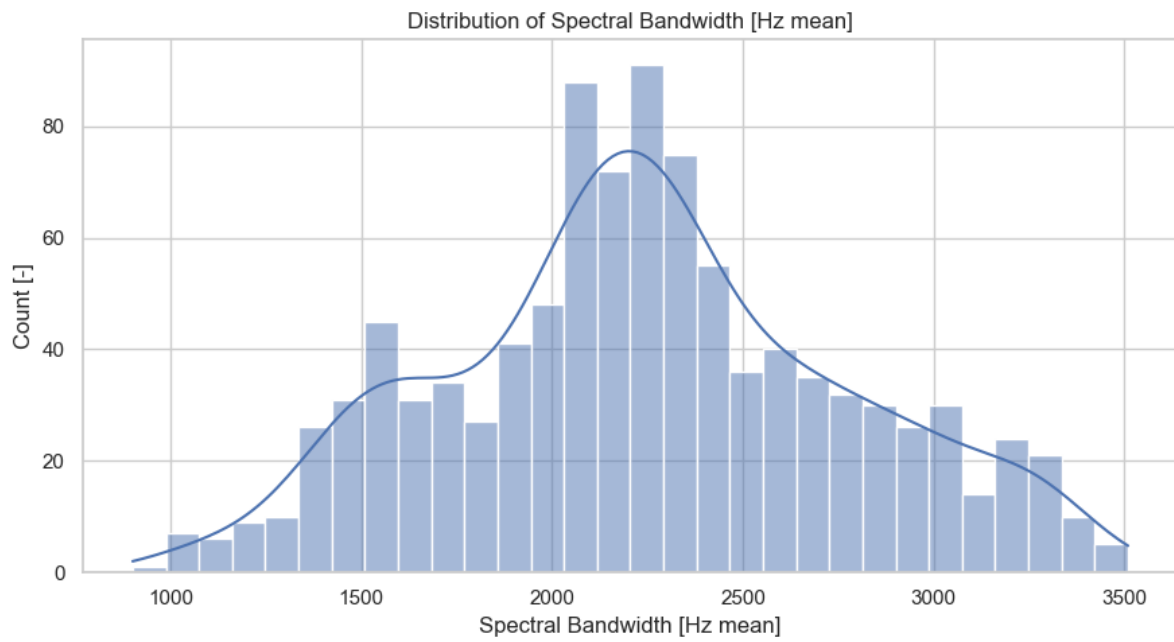


Figure 7 Distribution of Spectral Bandwidth (mean)

Spectral bandwidth describes the spread of frequencies around the spectral centroid.

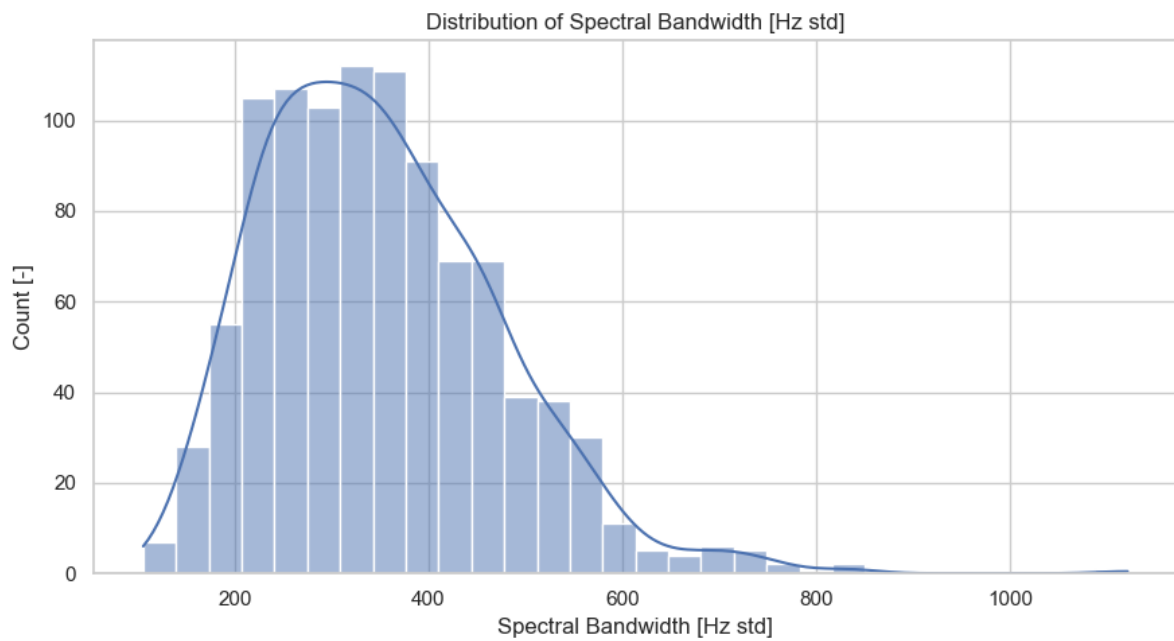


Figure 8 Distribution of Spectral Bandwidth (std)

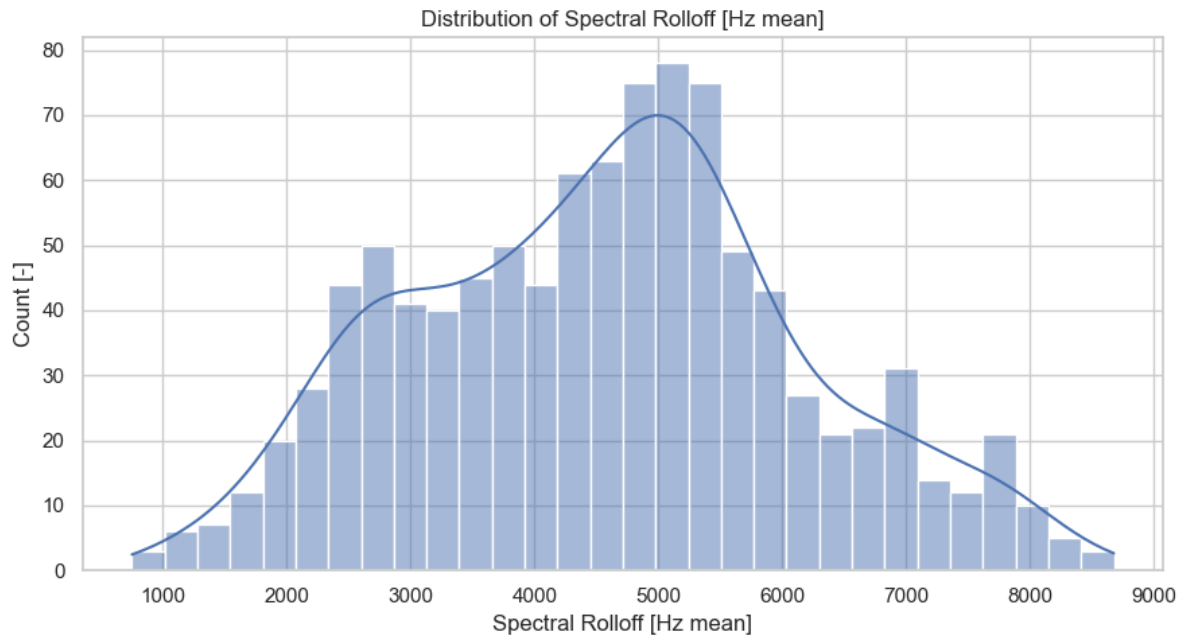


Figure 9 Distribution of Spectral Rolloff (mean)

Spectral rolloff measures the frequency below which most spectral energy is concentrated.

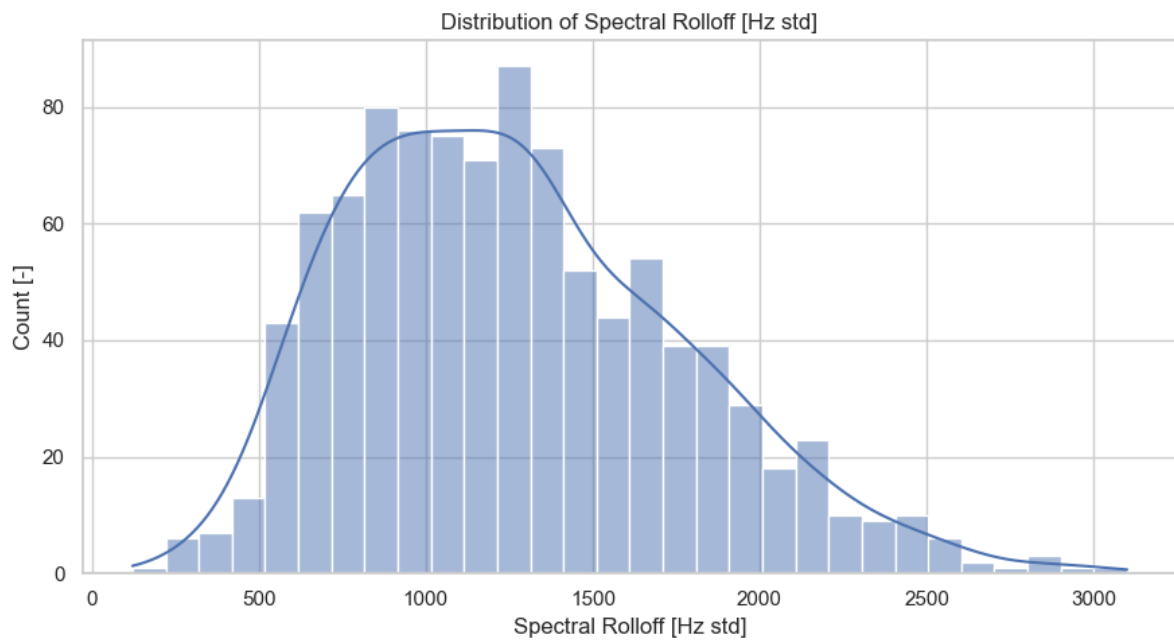


Figure 10 Figure 5.10 – Distribution of Spectral Rolloff (std)

5.6 Zero Crossing Rate Analysis

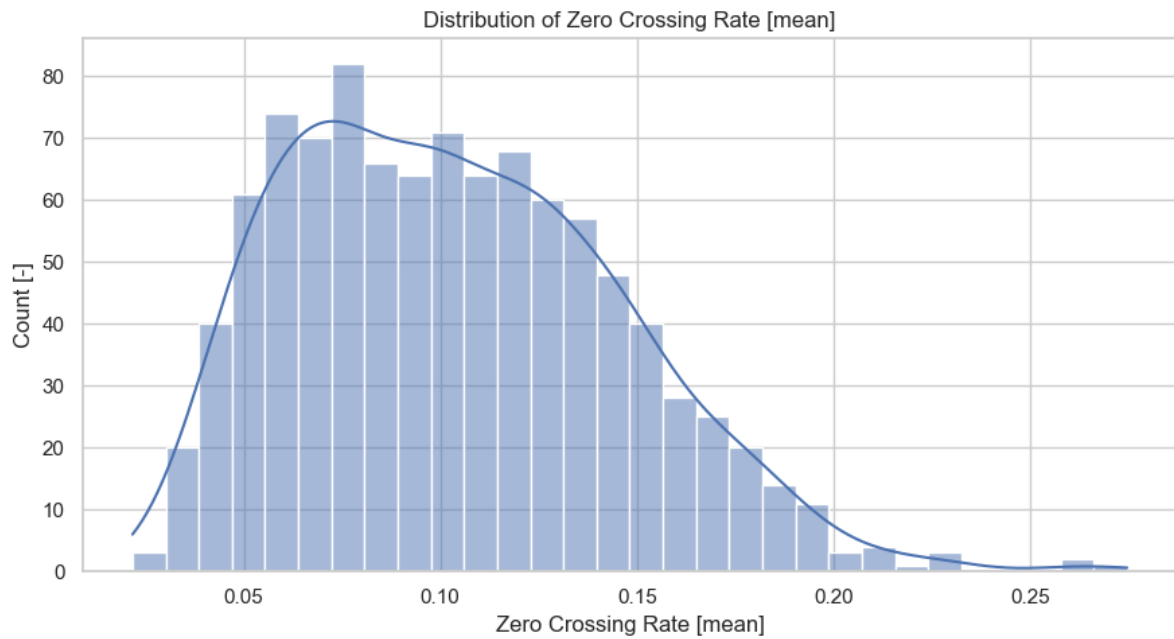


Figure 11 Distribution of Zero Crossing Rate (mean)

Zero crossing rate describes how frequently the signal crosses the zero amplitude axis. Higher values typically correspond to noisier signals or signals with strong percussive elements.

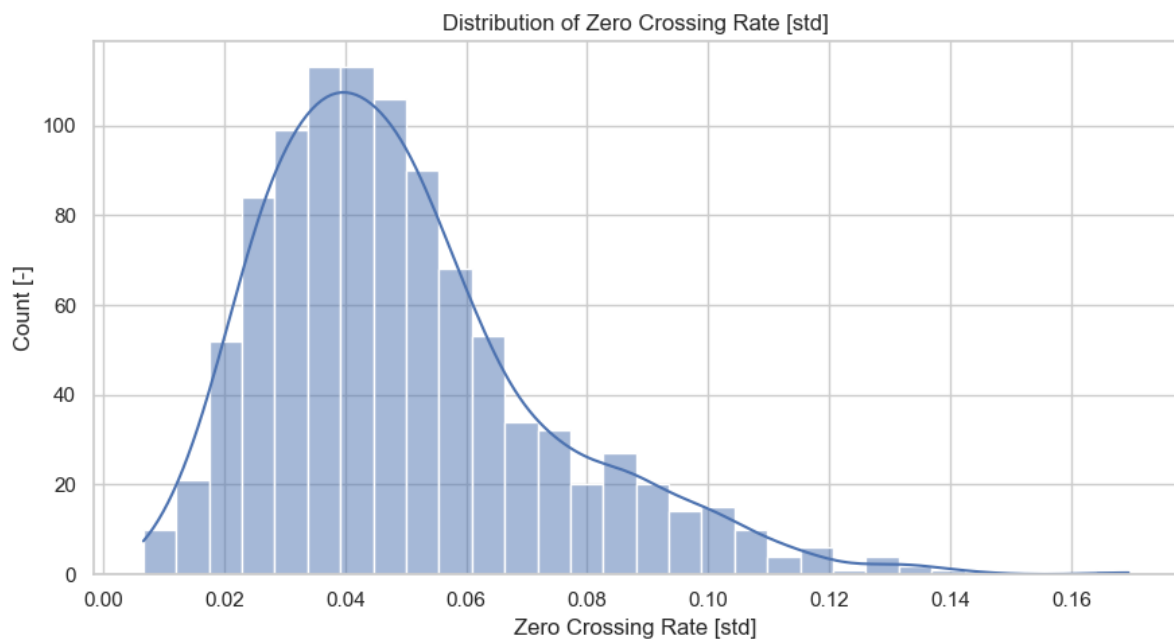


Figure 12 Distribution of Zero Crossing Rate (std)

5.7 MFCC Feature Analysis

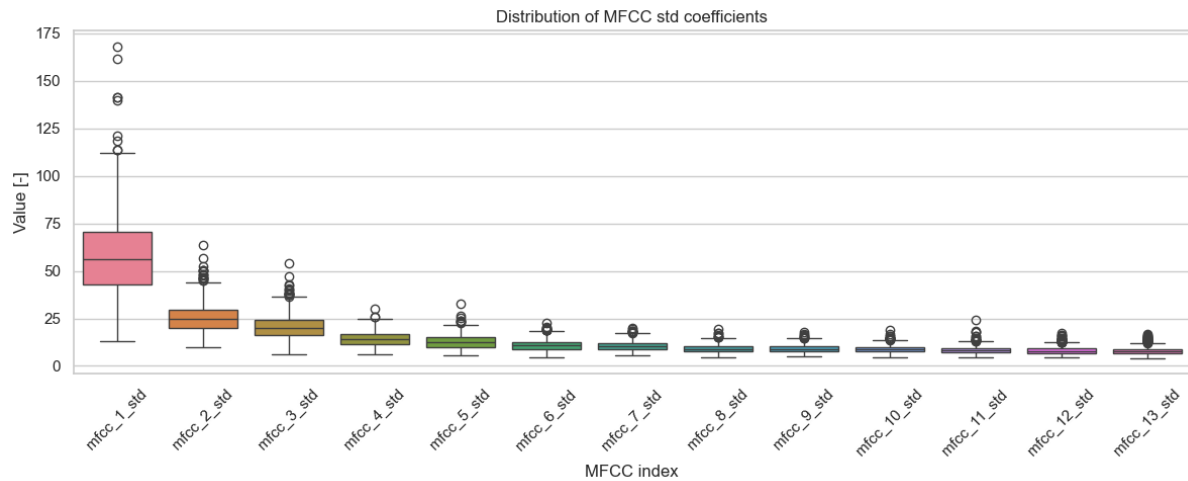


Figure 13 Distribution of MFCC mean coefficients

MFCC features capture perceptual characteristics of sound and are widely used in speech and music recognition systems.

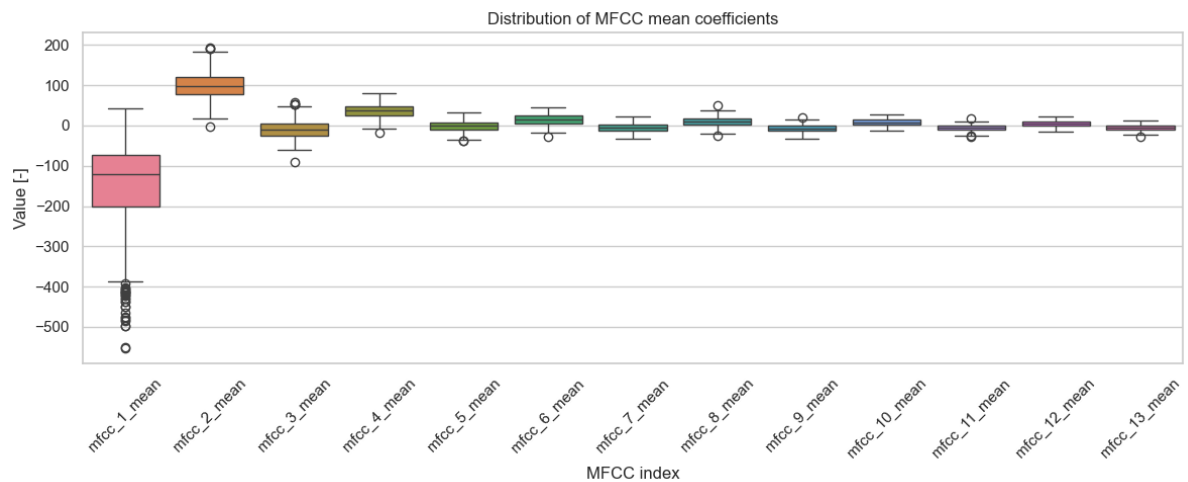


Figure 14 Distribution of MFCC standard deviation coefficients

The first MFCC coefficients exhibit larger variance compared to higher-order coefficients. This is expected because lower-order coefficients represent the most significant spectral envelope characteristics.

5.8 Chroma Feature Analysis

Chroma features represent tonal properties of the music signal.

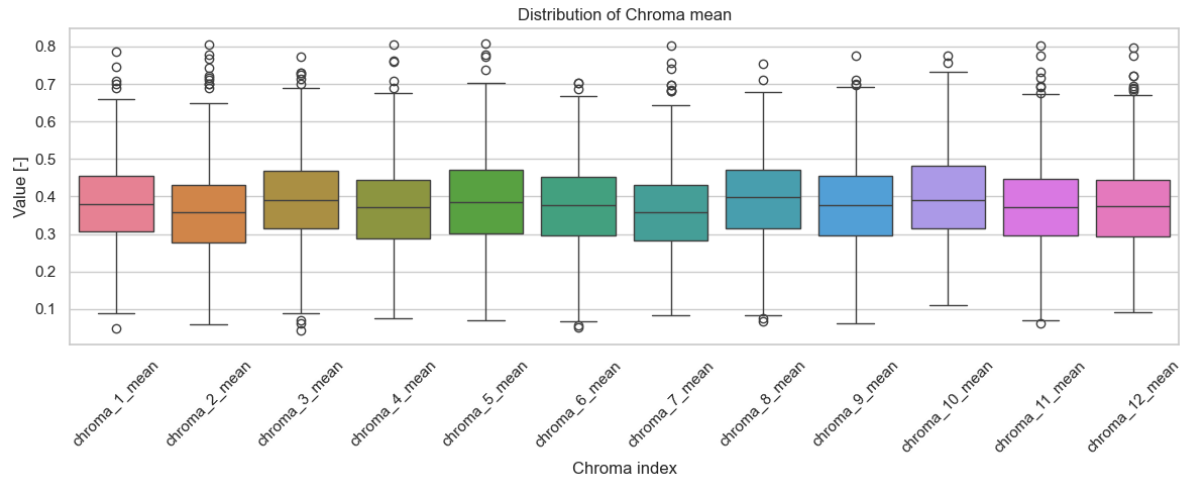


Figure 15 Distribution of Chroma mean values

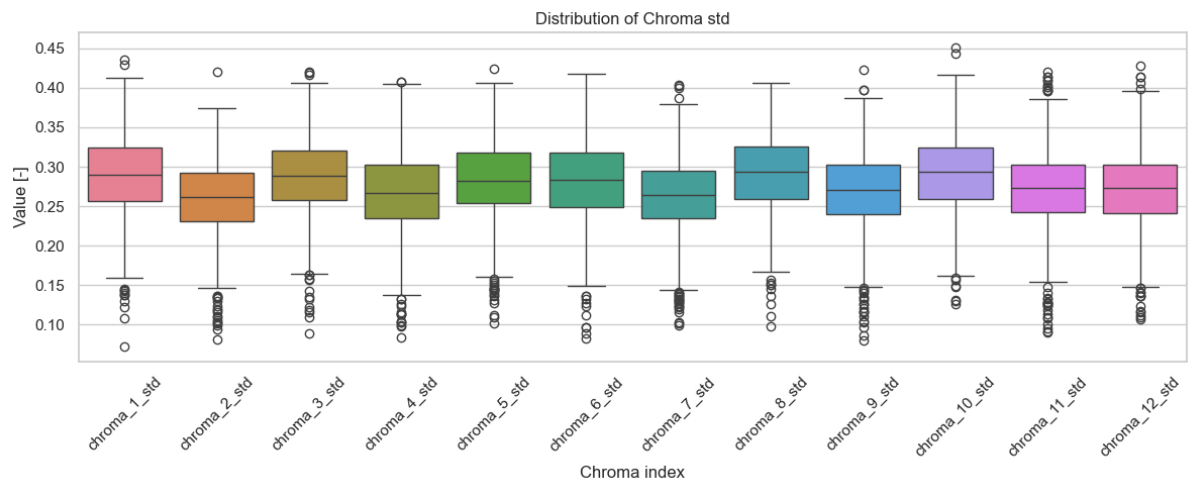


Figure 16 Distribution of Chroma standard deviation values

5.9 Feature Correlation Analysis

To examine relationships between the extracted audio features, a correlation matrix was computed.

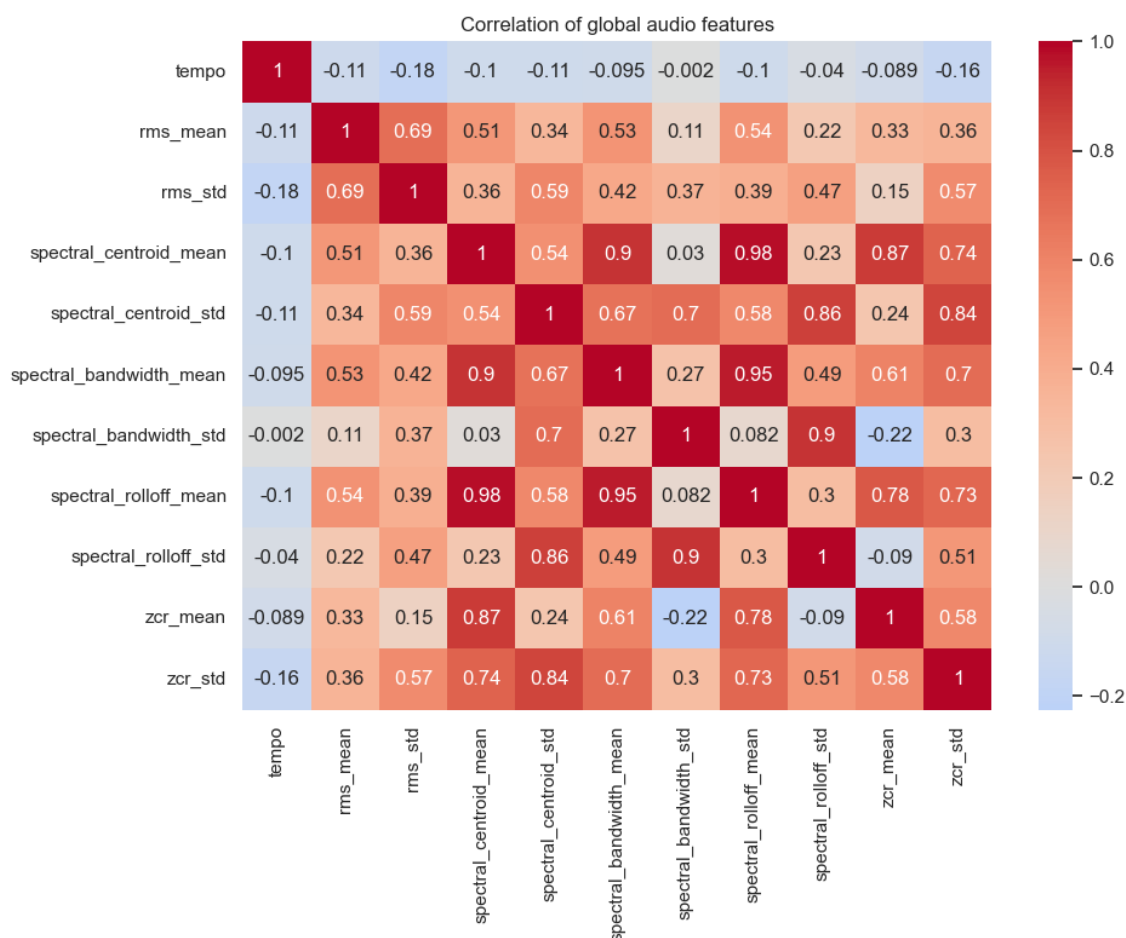


Figure 17 Correlation matrix of global audio features

The correlation heatmap reveals moderate correlations between several spectral features, particularly spectral centroid, spectral bandwidth, and spectral rolloff. These correlations are expected because these descriptors measure related aspects of the frequency distribution of the signal.

However, no extreme multicollinearity was observed, indicating that the features provide complementary information.

5.10 PCA Visualization

Principal Component Analysis (PCA) was used to visualize the structure of the dataset in a reduced-dimensional space.

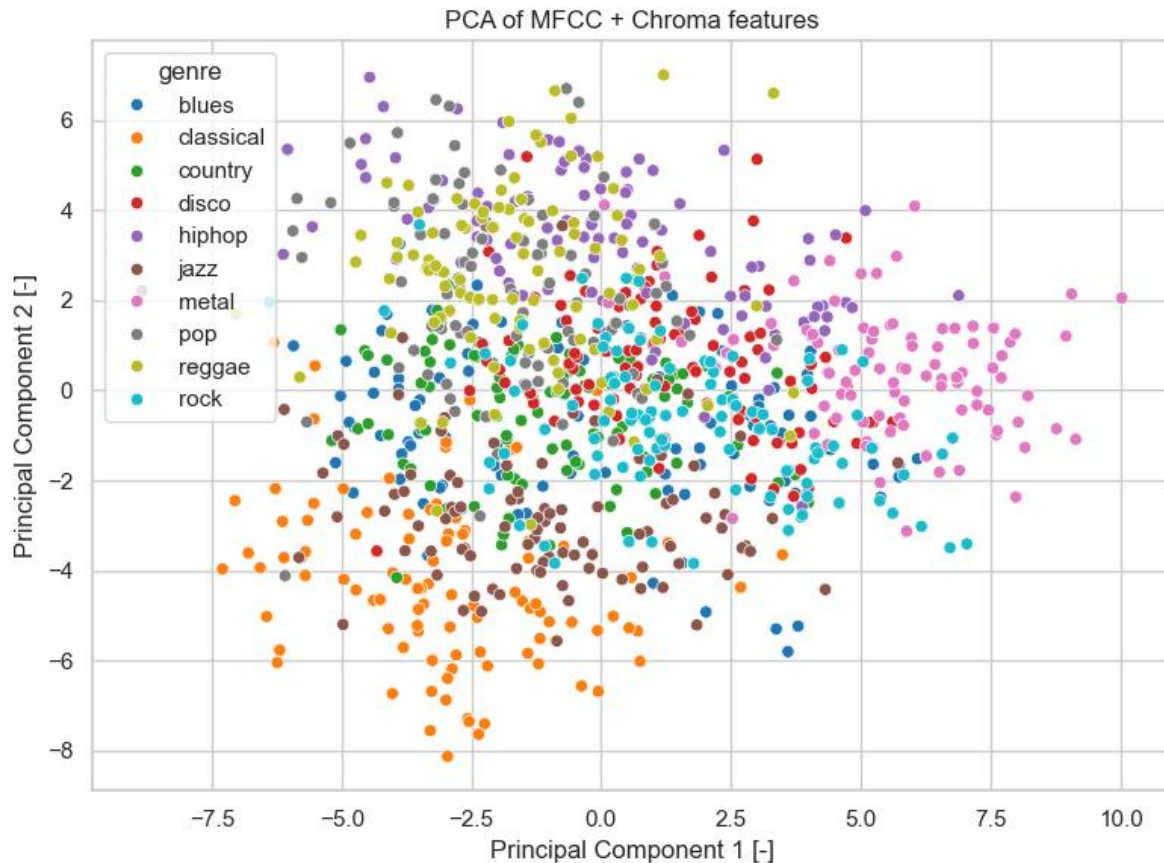


Figure 18 PCA projection of MFCC and chroma features

The PCA visualization shows partial clustering of musical genres. While the genres are not perfectly separable in two dimensions, several groups show tendencies to cluster together.

This suggests that the extracted audio features contain meaningful information for distinguishing between genres, although the classification boundary is likely non-linear.

5.11 Implications for Experimental Design

The results of the exploratory data analysis provide important insights for the subsequent machine learning experiments.

First, the wide range of feature values indicates the need for feature standardization. This observation directly justifies the use of `StandardScaler` in the preprocessing pipeline.

Second, the PCA visualization suggests that genre classes are not linearly separable, indicating that non-linear models such as Support Vector Machines may perform better than simple linear classifiers.

Third, the correlation analysis indicates some redundancy between spectral features, although it is not severe enough to require aggressive feature elimination.

Finally, the balanced class distribution ensures that standard classification metrics such as accuracy, precision, recall, and F1-score can be reliably used for model evaluation.

These findings support the selection of Support Vector Machines with non-linear kernels in subsequent experiments.

5.12 Dataset Bias and Limitations

Despite its usefulness as a benchmark dataset, the GTZAN dataset has several known limitations that must be considered when interpreting experimental results.

First, the dataset may contain duplicate recordings and artist overlap between samples. This introduces a potential data leakage risk, where highly similar recordings may appear in both training and testing subsets, artificially inflating classification performance.

Second, some tracks may contain incorrect or ambiguous genre labels. Since musical genres are often subjective and overlapping, the target labels may not always reflect clear acoustic boundaries. This introduces potential labeling noise into the classification task.

Third, the dataset contains recordings collected under different recording conditions, including radio broadcasts, CD recordings, and microphone captures. As a result, some extracted features may partially reflect recording artifacts rather than purely musical characteristics.

Additionally, the GTZAN dataset represents only ten predefined genres and therefore does not fully capture the diversity and complexity of real-world music distributions.

These limitations directly influence the experimental design used in subsequent assignments. In particular:

- stratified train-test splitting is required to preserve class balance,
- cross-validation will be used to reduce evaluation variance,
- preprocessing steps must be fitted only on training data to avoid information leakage,
- model performance will be interpreted cautiously due to possible dataset bias.

Therefore, the results obtained in this study should not be interpreted as universally generalizable to all music classification tasks, but rather as controlled experimental results on a standardized benchmark dataset.